TAYLER ERBE / DECISION BRIEF · OPEN-WEIGHT DOCUMENT AI

Which model should read your documents, and what will it cost to run?

Research proposals, procurement packets, contracts and reports keep their most useful facts in tables, charts, scans and clauses. I tested eleven open-weight models on public stand-ins for that work, confirmed the leaders on data they had never seen, and priced the hardware to run them privately.

> **Short answer:** use small specialised models, not one big one. The recommended document readers fit on one 24 GB data-centre GPU. Keep a reviewer on amounts and contract answers.

Tayler Erbe · Data scientist, applied AI · model evaluations that end in a decision: which model, on what hardware, at what cost.
About

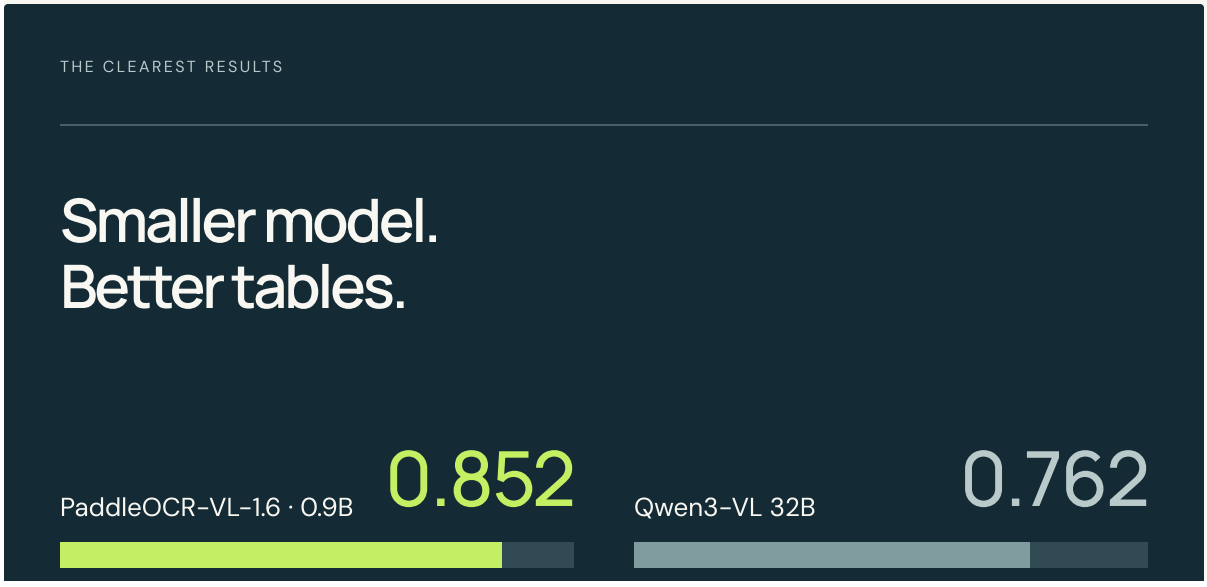
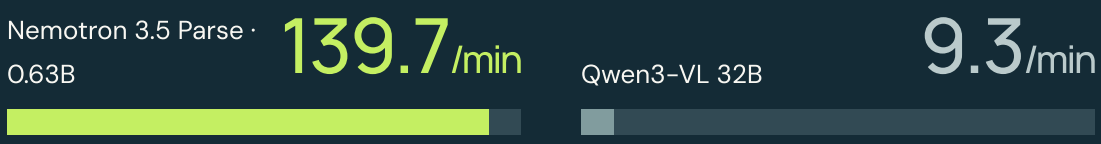


Table similarity on 100 unseen financial-report tables, same output limit. It also kept more numbers: 97.4% against 93.2%. Passed a test written down before scoring.

See the table comparison ↗

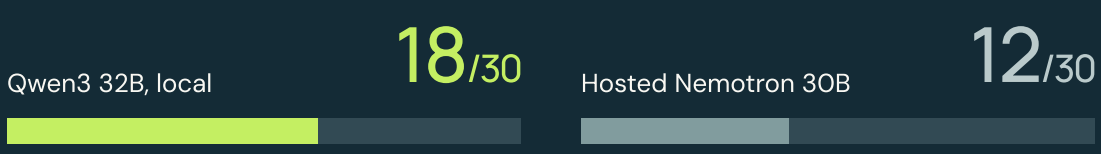
15 times faster on whole pages.



Items processed per minute on one batch of 20 pages and 20 tables, four requests at a time, same desktop GPU. Its page text was also more accurate (word overlap 0.899 against 0.820).

See the page comparison ↗

Local models led on contract questions.



ContractNLI statements on the same documents. On CUAD no configuration beat the 15 of 30 scored by always answering "absent", so every contract answer still needs review.

See the contract comparison ↗

0.9B parameter table reader that beat a 32B model	**$6k–10k** L4-class server that runs the document readers
15× faster page reading than the 32B model	**0** scores decided by another model

01 / FINDINGS

What to use for each job.

Each card names the job, the model, why it was chosen, the evidence and the main limit, and links to the comparison behind it.

BUDGET AND FINANCIAL TABLES

PaddleOCR-VL-1.6

Rebuilt financial tables more faithfully than a 32B vision model, and kept the numbers.

0.852
table similarity
vs 0.762 for 32B

97.4%
numbers recovered
vs 93.2%

Confirmed · pilot first

Below a 99% bar on numbers, so amounts still need review.

See the table comparison ↗

WHOLE-PAGE TEXT

Nemotron 3.5 Parse

The most accurate page text of the models tested, and the fastest.

0.899
word overlap
vs 0.820 for 32B

140/min
items processed
vs 9 for 32B

Lead · check numbers

Numbers matched the 32B on average (88.2% vs 88.3%), but 80 pages could not rule out a small loss.

See the page comparison ↗

CHARTS

Qwen3-VL, method per model

The same instruction helped one model and cut another's accuracy by about 43 points.

84.6%
32B with 3-answer voting, 1,322 questions

75.0%
8B at baseline (32.3% with self-check)

Method matters

On 24 GB or less use the 8B at baseline; with 48 GB, the 32B with voting.

See the chart comparison ↗

CONTRACT CLAUSES

Qwen3 32B · gpt-oss-20b

Best on contract questions that ask whether a document supports a statement.

18/30
ContractNLI, Qwen3 32B vs 14 gpt-oss, 12 hosted

15/30
CUAD "always absent" baseline: none beat it

Human review required

gpt-oss-20b is the option that fits a 24 GB card. Modest scores on 30 questions per task.

See the contract comparison ↗

CROPPED TEXT REGIONS

Qwen3-VL 8B

Read cropped paragraphs as accurately as the 32B models, on a 12 GB laptop GPU.

1.2%

character error
on 50 crops

Small is enough

Exploratory set of 50 crops.

See the text-region comparison ↗

FINDING TABLES AND FIGURES

DocLayout-YOLO

A dedicated layout detector found figures far more reliably than the page parser.

0.876

figure finding
vs 0.612 for the parser

Keep the detector

For tables the two were not clearly different.

See the layout comparison ↗

The top score is not always the recommendation.

The PaddleOCR official pipeline had the best word overlap on pages (0.922) but recovered only 77.1% of the numbers. These recommendations favour keeping numbers, because budgets, awards and invoices fail on numbers first.

Compare them ↗

02 / HARDWARE AND BUDGET

Start from the budget.

Pick a tier to see what it can run, whether that was measured or estimated, and a dated price for a complete system. Prices were checked in October 2026 and move quickly.

SINGLE NVIDIA L4 SERVER

$6,000 to $10,000

24 GB data-centre card, low power

Estimated fit · not benchmarked

What it runs

- Both recommended parsers, with room to spare
- Qwen3-VL 8B for charts and text regions
- gpt-oss-20b for contract questions

What that means

Covers every document-reading route. Any 32B model is tight.

Fit from weight sizes. Dated planning allowance, not a quote.

L4 OR L40S?

The answer depends on contracts.



The L4 covers the document-reading routes: tables, pages, charts, text regions and layout. The L40S mainly buys the 32B contract model and headroom. On the public contract questions the 32B led the 20B on ContractNLI (18 against 14 of 30) and they tied on CUAD. Whether that gap is worth about $10,000 more depends on your contract fields and the volume you need to process. A short pilot can measure both before you buy.

Speeds were measured on the 32 GB desktop. They compare models fairly but are not speeds for an L4 or L40S. Purchase price leaves out engineering, review time, maintenance and energy. Running models on your own hardware keeps documents in your environment, but it does not make the system secure by itself. Hardware and budget guide ↗

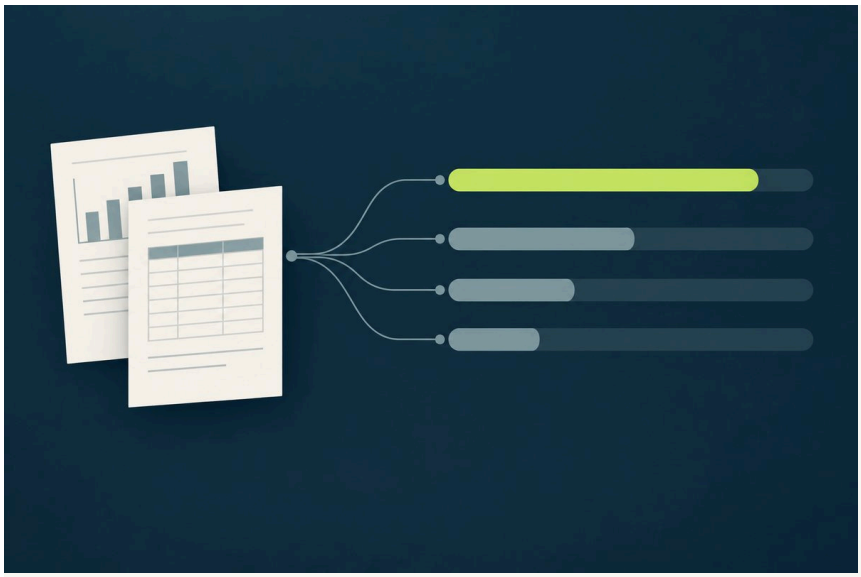
TIER	PRICE	RUNS	BASIS
Existing computer, no GPU	$0 extra	Read text and spreadsheets directly when the original file exists; Basic OCR and the review screens	Not benchmarked. A starting point for native files, not a scanned-document solution.

TIER	PRICE	RUNS	BASIS
16 GB GPU workstation	$1,500 to $2,500	Either recommended parser (about 1 to 2 GB of weights); Qwen3-VL 8B for charts and text regions (about 10 GB)	Estimated fit. Fit from weight sizes; not tested. Planning allowance from the hardware guide.
12 GB mobile workstation	About $4,650 new	Qwen3-VL 8B and Qwen2.5-VL 7B on tables, charts, chart-to-table, text regions and whole pages, one model at a time	Measured. Retail list price for a comparable configuration, October 2026. Refurbished units cost far less.
32 GB desktop, RTX 5090	About $3,600 to $5,300	All parsers and vision models, the 32B models and gpt-oss-20b; The two recommended parsers: 73 to 140 items a minute; the 32B vision model: 9	Measured. Complete prebuilt systems listed in 2026. Graphics-card shortages move these prices.
Single NVIDIA L4 server	$6,000 to $10,000	Both recommended parsers, with room to spare; Qwen3-VL 8B for charts and text regions; gpt-oss-20b for contract questions	Estimated fit · not benchmarked. Fit from weight sizes. Dated planning allowance, not a quote.
Single NVIDIA L40S server	$16,000 to $21,000	Everything recommended, including Qwen3 32B for contract questions; Qwen3-VL 32B with voting for charts; Headroom for several requests at once	Estimated fit · not benchmarked. Fit from weight sizes. Dated planning allowance, not a quote.

03 / RESULTS

The clearest results.

Three comparisons carry the recommendations. Every comparison, with its question, sample, measures and limits, is in the interactive explorer.



FINANCIAL TABLES · 100 UNSEEN

A 0.9B reader beat the 32B

PaddleOCR-VL-1.6 · table similarity 0.852

Qwen3-VL 32B 0.762

PaddleOCR-VL-1.6 · numbers recovered 97.4%

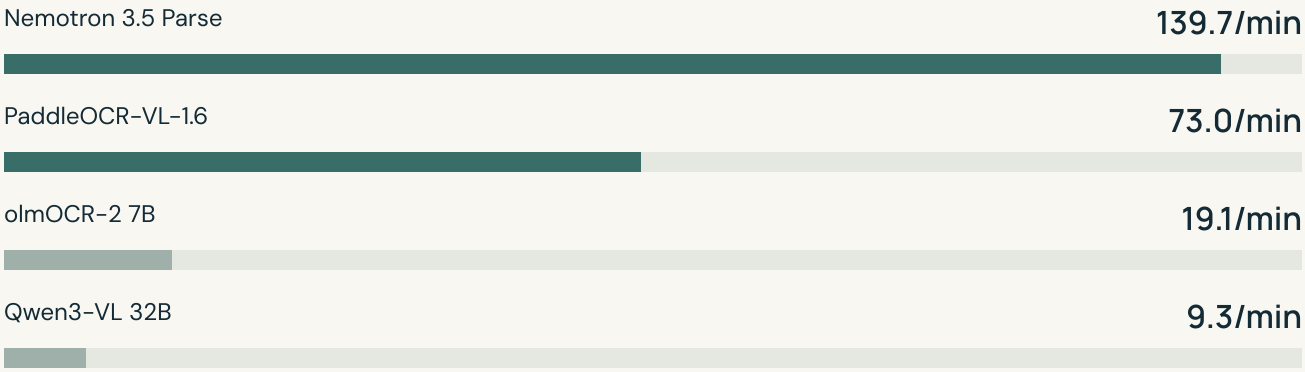
Qwen3-VL 32B 93.2%

Passed the pre-registered test. Similarity measures table structure, not fully correct tables.

Open the comparison ↗

SPEED · ONE BATCH, SAME DESKTOP

Small parsers are much faster

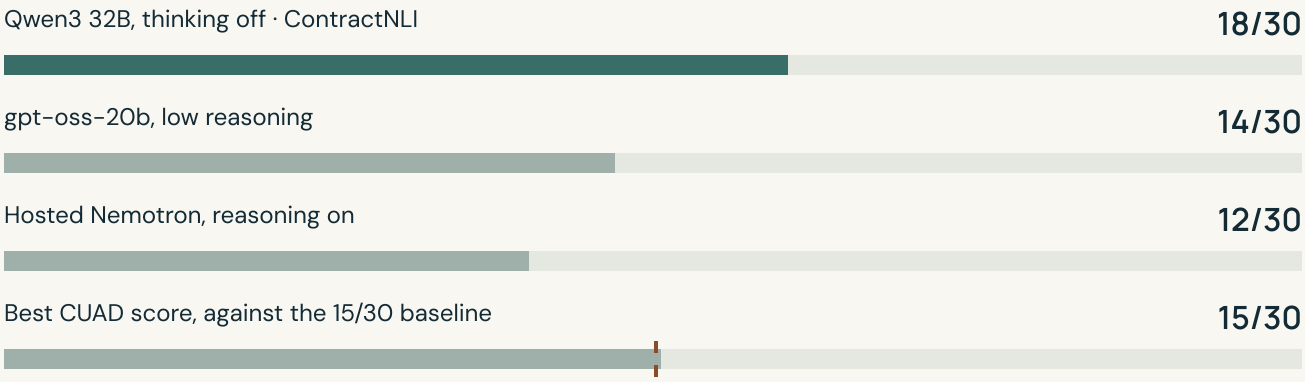


Relative speeds on one machine, not a promise for other hardware.

Open the page comparison ↗

CONTRACTS · SAME 60 DOCUMENTS

Qwen3 32B led ContractNLI



CUAD does not separate these models. Answers need review.

Open the contract comparison ↗

Explore all 18 comparisons ↗

04 / METHOD

How the decision was made, and how I would make it for you.

The method matters more than any single score. It is the same six steps whether the documents are research awards, procurement packets or contracts.

01

Define the jobs

Split the documents into the kinds of reading they need, and name the failure that matters for each. Here, that was losing numbers.

02

Freeze fair tests

Pick public stand-ins with answer keys, fix the questions and the scoring code, and let no model grade another.

03

Shortlist by budget

Choose candidates that fit the hardware you can afford, from small parsers to 32B models.

04

Explore, then confirm

Explore on development data, write down what will count as a win, then test once on data no model has seen.

05

Price the fit

Map the winning setup to a hardware tier and a dated budget, separating measured speed from estimates.

06

Validate on real documents

Label a representative set of your own documents and measure field accuracy before approving anything.

Why it took several rounds

Each round answered a question the previous one raised. None simply reran tests until a good number appeared. Pick a round.

QUESTION

Which local vision model reads each part of a document, on a 12 GB laptop and a 32 GB desktop?

TEST

Five vision models and several prompting methods on frozen public sets for text, tables, charts and pages.

FINDING

The 8B model matched the 32B on text and tables. Prompting methods helped some models and hurt others.

DECISION

Route by model and task. Test the promising methods properly.

QUESTION

Do the chosen methods hold on data no model has seen?

TEST

Three-answer voting on 1,322 new chart questions and layout masking on 80 new pages, with the pass rules written first.

FINDING

Voting helped the 7B and 32B, not the 8B. Masking cleaned up text but lost numbers.

DECISION

Keep voting optional, keep masking experimental, and look for specialist tools.

QUESTION

Can models answer contract questions with evidence a reviewer can check?

TEST

CUAD and ContractNLI public contracts, quoting versus selecting evidence by ID, and a controlled reassessment of a hosted model.

FINDING

Selecting evidence by ID removed copy failures. The hosted model's poor first score was a configuration problem.

DECISION

Compare every candidate on the same 60 documents.

QUESTION

Do specialist document parsers or gpt-oss-20b beat the incumbents?

TEST

Three parsers on every earlier page and table set, and gpt-oss-20b on the contract sets.

FINDING

Leads emerged: PaddleOCR-VL for tables, Nemotron Parse for pages. They were found by looking, so they were not yet proof.

DECISION

Write the two hypotheses down and test them on fresh data.

QUESTION

Do the leads hold on unseen data, with the same output limit for every model?

TEST

100 new financial tables and 80 new pages, the parsers' official pipelines, budget checks, and Qwen3 32B on the 60 contract documents. 3c reran two arms that had failed for software reasons.

FINDING

The table lead was confirmed. The page lead held on text but was inconclusive on numbers. Official pipelines did not beat a single pass.

DECISION

Close the evaluation, route by task, and validate on real documents.

Why it stopped: the leads were either confirmed or bounded. More public benchmarks would add little, so the next evidence has to come from real documents. Full methods and definitions ↗

From evaluation to a working service.

A mixed PDF is not one task. The proposed service sends each part of a document to the tool that did best on that kind of reading, keeps every answer linked to its source page, and leaves the decision with a reviewer.



Receive

Keep the original file untouched.

Identify

Find pages, tables, charts and text. Use native text where it exists.

Route

Send each part to its recommended model.

Combine

Assemble one record, linked to source pages.

Review

Show values beside the page. Flag numbers and unread parts.

Export

Send approved data to search, reports or other systems.

01 / DEFINE

Agree the fields

Which fields matter, where the documents come from, and where approved data goes.

02 / LABEL

Build a review set

About 100 representative documents, including difficult scans, with two reviewers on a subset.

03 / MEASURE

Run the routes

Field accuracy, numbers, traceability, processing time and correction effort on the target server.

04 / DECIDE

Scale or adapt

Scale, adjust retrieval or prompts, or fine-tune a model such as Qwen3-VL 8B, based on the measured errors.

Clearing a backlog vs keeping up. A first pass over an archive can run in batches: load one model, process every waiting table, then switch, so one GPU never holds everything at once. After that, new documents arrive in smaller volumes. How long a backlog takes cannot be estimated from these tests; the pilot measures it.

Implementation roadmap ↗

05 / REPORTS AND DATA

Read it, share it, check it.

Reports open as web pages, each with a matching PDF. Data and code follow for anyone who wants to check the numbers.

Final advisory · recommendations, confirmed tests, limits ↗

Final advisory · PDF ↓

Hardware and budget guide ↗

Hardware and budget guide · PDF ↓

Implementation roadmap ↗

Hosted model reassessment ↗

Routing report, rounds 1 and 2 (historical) ↗

Every comparison on this site · JSON ↓

Final aggregates with checksums · JSON ↓

Code, frozen protocols and tests · GitHub ↗

ABOUT THIS WORK

Tayler Erbe

I am a data scientist working in applied AI. I design evaluations that turn model results into decisions: which model to use for each job, what hardware it needs, what it costs, and what still has to be proven.

This study used public benchmarks only. Local models did all of the document reading and answering. AI coding assistants helped build the test harness, and no model graded another. To talk about an evaluation for your own documents, see the case study on my site or get in touch.

Document Intelligence Lab

Tayler Erbe · Last updated 29 October 2026 · Public benchmarks only. No private documents, labels or model responses.

Full evidence and methods ↗